

Master's Report — proposed tables

all-numbers dump, 2026-05-28

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1 Table 1 — Pre-trade-off snapshot at step 400K

Same layout as Table 2 (step 700K), but at step **400K** — well before the T-D crossover ($\sim 850K$ for PDB). PDB: 3 seeds each. AFDB L9-GN and L4-MPNN: 1 seed only. Green = improvement; red = regression.

Variant	Quality		T-W			T-D		S-W	S-D	
	Des $\uparrow$	FID $\downarrow$	fJSD-A $\downarrow$	fS-A $\uparrow$	pwTM $\downarrow$	#Clust $\uparrow$	pwTM $\downarrow$	ssJSD2D $\downarrow$	ssJSD2D $\downarrow$	E%des $\uparrow$
<i>PDB-trained, n=256, 400K, 3 seeds; FID is FID-PDB (in-distribution).</i>										
Baseline	0.26	472	0.52	5.64	0.14	54	0.14	0.27	0.55	0.07
REPA L4-random	+0.13	+34	+0.35	+0.30	+0.04	+1	+0.03	−0.07	−0.15	+0.04
REPA L4-GN	+0.26	−32	+0.28	+1.8	+0.08	+18	+0.12	−0.07	−0.15	+0.09
REPA L9-GN	+0.26	−132	+0.01	+1.3	+0.02	+38	+0.03	−0.06	−0.12	+0.07
REPA L4-MPNN	+0.33	+40	+0.12	+0.50	+0.13	+30	+0.18	−0.01	−0.20	+0.18
REPA L9-MPNN	+0.39	−62	+0.30	−0.04	+0.03	+40	+0.03	+0.06	−0.15	+0.10
<i>AFDB-trained, n=256, 400K, 3 seeds (L9-GN, L4-MPNN: 1 seed); FID is FID-AFDB.</i>										
Baseline	0.75	431	1.44	6.46	0.18	157	0.17	0.14	0.23	0.17
REPA L4-GN	+0.03	−103	−0.15	+0.65	+0.00	−34	+0.04	−0.10	−0.10	−0.03
REPA L9-GN	−0.03	−119	−0.32	+0.52	—	−61	+0.14	−0.09	−0.14	+0.03
REPA L4-MPNN	−0.07	+45	+0.51	−1.5	—	−62	+0.11	−0.02	−0.11	−0.01
REPA L9-MPNN	−0.05	−79	+0.23	−1.7	−0.01	−20	+0.01	−0.00	−0.07	−0.03

Table 1: 400K snapshot: pre-trade-off. At this early step, **all REPA variants are net-positive on T-D #Clust on PDB** (+1 to +40), and GearNet consistently wins FID/fS-A while MPNN leads on Des. The T-W fJSD-A regressions (most variants red at PDB) reflect early training noise at low step counts rather than a structural failure. AFDB already shows L4-GN as the clear FID leader (−103), but MPNN variants are still converging (−79 for L9-MPNN). The T-D collapse seen at 1M has not yet appeared: #Clust is uniformly positive on PDB, and AFDB L9-MPNN is already showing positive #Clust trajectory.

2 Table 2 — Centerpiece: all metrics $\times$ encoder at step 700K

Multi-seed mean (3 seeds for most cells; MPNN-L4-AFDB is 1 seed). **Baseline** row gives absolute values; REPA rows give Δ (**REPA** − **baseline**). Arrows on column headers indicate the direction of improvement. Green = improvement over baseline; red = regression.

Variant	Quality		T-W			T-D		S-W	S-D	
	Des $\uparrow$	FID $\downarrow$	fJSD-A $\downarrow$	fS-A $\uparrow$	pwTM $\downarrow$	#Clust $\uparrow$	pwTM $\downarrow$	ssJSD2D $\downarrow$	ssJSD2D $\downarrow$	E%des $\uparrow$
<i>PDB-trained, n=256, 700K, 3 seeds; FID is FID-PDB (in-distribution).</i>										
Baseline	0.46	437	1.11	5.48	0.22	77	0.27	0.35	0.40	0.22
REPA L4-random	-0.08	+34	-0.55	+0.05	-0.05	-23	-0.10	-0.13	-0.01	-0.11
REPA L4-GN	+0.24	+71	-0.38	+1.63	+0.13	+1	+0.11	-0.21	-0.12	+0.00
REPA L9-GN	+0.14	-118	-0.57	+2.31	-0.03	+27	-0.02	-0.19	-0.10	-0.04
REPA L4-MPNN	+0.18	-88	-0.13	+0.48	+0.05	-4	+0.08	-0.16	-0.14	+0.01
REPA L9-MPNN	+0.05	-19	-0.32	+0.37	-0.01	+33	-0.01	-0.14	-0.11	-0.07
<i>AFDB-trained, n=256, 700K, 3 seeds (MPNN-L4: 1 seed); FID is FID-AFDB.</i>										
Baseline	0.72	494	2.45	4.59	0.20	128	0.22	0.15	0.16	0.15
REPA L4-GN	+0.05	-195	-1.56	+2.27	+0.00	-35	-0.04	-0.11	-0.07	-0.02
REPA L9-GN	+0.01	-172	-1.29	+2.76	+0.01	-50	+0.11	-0.12	-0.06	+0.04
REPA L4-MPNN	-0.03	-22	-0.39	+0.03	—	-1	-0.07	-0.02	+0.05	-0.01
REPA L9-MPNN	+0.05	-139	-0.93	+0.31	-0.03	+21	-0.03	+0.01	+0.02	-0.04

Table 2: The encoder-routing pattern at one fixed step. GearNet $\rightarrow$ best T-W (FID, fJSD-A, fS-A) on both datasets. MPNN $\rightarrow$ best quality (Des). T-D: GearNet reduces designable diversity on AFDB (-35, -50 #Clust); MPNN-L9-AFDB preserves it (+21, the clean falsifier). Random control (PDB only): wins on T-W and S-W distribution-match metrics (fJSD-A, ssJSD2D, pwTM) but *hurts* quality (Des -0.08) and designable diversity (#Clust -23, E%des -0.11) — a regulariser without quality benefit.

3 Table 3 — Post-trade-off snapshot at step 1.0M

Same metrics as Table 2, snapshot at **step 1.0M** — past the PDB T-D crossover ($\sim 850K$) and well past the AFDB crossover ($\sim 150K$). PDB L4-random not run past 700K (—); AFDB L9-GearNet stops at 900K (—); AFDB L4-MPNN whole-set pwTM unavailable. Green = improvement; red = regression.

Variant	Quality		T-W			T-D		S-W	S-D	
	Des $\uparrow$	FID $\downarrow$	fJSD-A $\downarrow$	fS-A $\uparrow$	pwTM $\downarrow$	#Clust $\uparrow$	pwTM $\downarrow$	ssJSD2D $\downarrow$	ssJSD2D $\downarrow$	E%des $\uparrow$
<i>PDB-trained, n=256, 1.0M, 3 seeds; FID is FID-PDB.</i>										
Baseline	0.54	467	0.95	5.67	0.17	114	0.16	0.32	0.39	0.15
REPA L4-GN	-0.09	-35	-0.41	+1.68	+0.00	-27	+0.01	-0.16	+0.00	-0.07
REPA L9-GN	-0.02	-130	+0.06	+0.92	+0.06	-44	+0.13	-0.19	-0.13	+0.09
REPA L4-MPNN	-0.20	+27	+0.10	-0.27	+0.00	-62	+0.07	-0.26	-0.26	+0.02
REPA L9-MPNN	+0.01	-81	-0.05	+0.29	+0.07	-51	+0.15	-0.15	-0.08	+0.04
REPA L4-random	—	—	—	—	—	—	—	—	—	—
<i>AFDB-trained, n=256, 1.0M, 3 seeds (MPNN-L4: 1 seed); FID is FID-AFDB.</i>										
Baseline	0.72	534	2.23	4.35	0.20	133	0.21	0.16	0.18	0.14
REPA L4-GN	+0.01	-228	-1.43	+3.68	-0.00	-39	+0.03	-0.12	-0.09	+0.01
REPA L9-GN	—	—	—	—	—	—	—	—	—	—
REPA L4-MPNN	-0.18	-118	-0.57	+1.20	—	-29	-0.06	-0.06	+0.09	+0.02
REPA L9-MPNN	-0.01	-101	-0.74	+0.44	-0.04	+6	-0.02	+0.02	+0.01	-0.03

Table 3: Post-trade-off: every PDB REPA variant now loses on T-D #Clust (-27 to -62). T-W gains persist (REPA still wins FID and fS-A on most variants) but tertiary diversity within the designable subset collapses. **AFDB L9-MPNN remains the falsifier** — the only REPA variant in the table with a positive #Clust Δ (+6), and the encoder $\times$ dataset combination that escapes the trade-off entirely. The contrast against Table 2’s 700K snapshot makes the temporal nature of the trade-off explicit: REPA’s T-W gain is durable, the T-D loss develops with training.

4 Table 4 — Speedup: REPA reaches the baseline’s best FID in $N \times$ fewer steps

Dataset	Metric	Variant	Baseline best	REPA reaches @	Speedup	REPA’s own best
<i>AFDB-GearNet — durable, clean:</i>						
AFDB	FID-AFDB	L4-GN	386 @1.3M	100K	13.0 $\times$	282
AFDB	FID-AFDB	L9-GN	386 @1.3M	200K	6.5 $\times$	313
AFDB	FID-AFDB	MPNN-L9	386 @1.3M	400K	3.2 $\times$	352
AFDB	FID-AFDB	MPNN-L4	386 @1.3M	never	—	416
<i>PDB — transient acceleration; baseline catches up late:</i>						
PDB	FID-PDB	L9-GN	$\sim$ 330 plateau	700K	$\sim$ 1.3–1.5 $\times$	319 (@1.2M)
PDB	FID-PDB	L9-MPNN	$\sim$ 330 plateau	1.1M	$\sim$ 1.3 $\times$	334 (@1.1M)
PDB	FID-PDB	MPNN-L4	329 plateau (full 1.6M)	1.6M	budget-matched	329 (@1.6M)

Table 4: Acceleration is durable on AFDB, transient on PDB. “Baseline best” = cumulative-min FID across baseline’s training; “REPA reaches” = earliest step REPA’s cumulative-min hits the baseline’s best. On AFDB, baseline FID-AFDB floors at 386 throughout 1.7M; REPA-GearNet both reaches that floor in 100–200K AND keeps improving past it (282–313). On PDB, REPA-L9 leads at every step $\leq$ 1.2M but the long-trained baseline matches and edges below by 1.6M (288). PDB framing: transient acceleration to REPA’s plateau, not a “reaches-baseline-final” speedup.

5 Table 5 — Sampler-noise robustness: Δ across γ

For the chapter’s best-PDB and best-AFDB REPA variants, mean Δ (REPA – baseline) across step-matched cells at each sampler noise level. Sign reflects effect direction; for “lower-is-better” metrics (FID, fJSD-A, ssJSD2D) a negative Δ is a REPA win. Green = improvement; red = regression.

Δ (REPA – baseline) ODE	$\gamma = 0$	$\gamma = 0.35$	$\gamma = 0.45$	$\gamma = 0.5$	$\gamma = 1.0$	
<i>PDB — REPA L9-MPNN vs Baseline (step-matched cells; baseline rows for $\gamma=0.35, 0.5$ recovered from older clean.jsonl).</i>						
Des $\uparrow$	+0.06	+0.07	+0.16	+0.18	+0.15	+0.04
FID $\downarrow$	–22	–1225	–78	–77	–89	–131
fJSD-A $\downarrow$	–0.13	+0.18	+0.21	–0.06	+0.07	–0.30
ssJSD2D $\downarrow$	–0.11	+0.08	–0.01	–0.03	–0.00	–0.10
#Clust $\uparrow$	+9	+8	–7	+4	+12	+2
<i>AFDB — REPA L4-GearNet vs Baseline (step-matched cells).</i>						
Des $\uparrow$	+0.16	+0.08	+0.07	+0.06	+0.06	+0.08
FID $\downarrow$	–102	–45	–133	–144	–132	–97
fJSD-A $\downarrow$	–0.31	+0.28	–0.94	–1.06	–0.92	–0.24
ssJSD2D $\downarrow$	–0.04	–0.19	–0.13	–0.13	–0.10	–0.03
#Clust $\uparrow$	+32	–15	–27	–23	–21	+10

Table 5: REPA’s encoder-routed effects are sampler-invariant in the middle band ($\gamma \in [0.35, 0.5]$) and largely survive at the extremes. AFDB shows the clean ODE designability floor (Δ Des = +0.16) and durable FID gains across all γ . PDB at $\gamma=0$ shows a striking –1225 FID gain: the PDB baseline at $\gamma=0$ mode-collapses (high Des, but very high FID); REPA’s distribution-match gain is correspondingly large. The earlier $\gamma=1$ distribution- match-collapse claim is retracted — L9-MPNN actually *wins* fJSD-A at $\gamma=1$ (–0.30), and only $\gamma=0$ shows the GearNet-specific Des-loss (not visible for MPNN here).

6 Table 6 — Representation rank-order (PDB-trained, best layer)

	IF top-1 $\uparrow$	dihedral MAE ($^{\circ}$) $\downarrow$	CATH-A $\uparrow$	CATH-T $\uparrow$
Baseline	0.123	32.1	0.407	0.301
REPA L4-random	0.146	20.2	0.476	0.378
REPA L9-GN	0.156	19.5	0.702	0.752
REPA L9-MPNN	0.167	16.4	0.548	0.473

Table 6: Encoder rank-order on student representation quality, best-layer, mean across steps $\geq 700K$, $n=256$ PDB. The axis split mirrors the gen-side encoder-routing (Table 2): MPNN-L9 wins per-residue probes (IF, dihedral); GearNet-L9 wins per-chain CATH probes. Random gives the smallest gain on every probe — learned target needed. CATH absolute values use the cleantrain protocol (highest n); IF and dihedral use xclean (doubly-clean cross-DB val).